

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

Detailed Research Paper and Project Documentation

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Prepared using: Excel, VS Code, Python, NumPy, Pandas, Matplotlib, and Streamlit

Project focus: Data cleaning, exploratory data analysis, route efficiency, logistics insights, and interactive dashboard development.

Abstract

This research paper presents a complete analytical approach for evaluating factory-to-customer shipping efficiency at Nassau Candy Distributor. The project converts raw order and shipment records into logistics intelligence by measuring shipping lead time, route volume, geographic performance, factory contribution, and ship mode behavior. The work combines spreadsheet review, Python programming, statistical preparation, visualization, and interactive web application development.

Because the available project documentation describes the analytical workflow but does not include the final aggregated numeric results, this paper explains the analysis framework and the business meaning of each measure without inventing unsupported values. The resulting process can be reused whenever the cleaned dataset is available.

Keywords

Logistics analytics; route efficiency; shipping lead time; supply chain; data cleaning; Streamlit dashboard; Nassau Candy Distributor.

1. Introduction and Business Context

Nassau Candy Distributor operates a nationwide distribution network in which products are manufactured at different factories and shipped to customers across the United States. Each order contains commercial, geographic, product, and shipment information. When these fields are studied together, they can reveal how quickly orders move, where delays occur, and which routes require operational attention.

Shipping efficiency is important for three connected reasons. First, delivery speed affects customer satisfaction and service reliability. Second, delays can increase handling, storage, communication, and transportation costs. Third, inefficient routes can limit the company's ability to scale as order volume grows. A route that appears acceptable at low volume may become a bottleneck when demand increases.

The central business problem is the lack of route-level visibility. Management needs to know which factory-to-customer routes perform well, which routes experience repeated delays, how results differ by region and state, and whether a particular shipping method provides better time performance. This project addresses those questions through a repeatable data analytics workflow.

Research Questions

- What is the typical shipping lead time from order placement to shipment?
- Which factory-to-state and factory-to-region routes are the fastest or slowest?
- Which locations combine high shipment volume with weak delivery performance?
- How does shipping performance vary by ship mode?
- How can an interactive dashboard support regular logistics decisions?

Project Objectives

- Prepare a reliable and analysis-ready shipping dataset.
- Create logistics features that describe time, factory, and route behavior.
- Measure route performance using volume, average lead time, variability, and delay frequency.
- Create visual and interactive outputs for business users.
- Translate analytical results into practical recommendations.

2. Dataset Description and Data Understanding

The dataset contains order-level records. Each row represents a transaction or shipment record and includes identifiers, dates, customer geography, product information, financial measures, and shipping attributes. Understanding the meaning of each field is essential because incorrect interpretation can lead to incorrect route rankings or misleading business conclusions.

Field	Business meaning
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date when the customer order was placed
Ship Date	Date when the order was shipped
Ship Mode	Shipping method used for the order
Customer ID	Unique customer identifier
City / State / Postal Code	Customer location
Division	Product category or division
Region	Customer region
Product ID / Product Name	Product identifiers and names
Sales	Revenue value of the order
Units	Quantity sold
Gross Profit	Sales minus manufacturing cost
Cost	Manufacturing cost

The project documentation also provides a product-to-factory mapping. Chocolate products such as Wonka Bar variants are assigned to Lot's O' Nuts or Wicked Choccy's; sugar products are assigned to Sugar Shack, Secret Factory, or The Other Factory; and selected "Other" products are mapped to Secret Factory or The Other Factory. This mapping is required because the source order data identifies a product, while route analysis requires the factory that fulfills it.

3. Data Cleaning and Validation

Data cleaning is the foundation of the project. Logistics analysis is highly sensitive to date errors, duplicated orders, inconsistent geography, and invalid financial values. A single negative lead time, for example, could incorrectly make a route appear exceptionally fast. Cleaning therefore converts the raw data into a consistent analytical table while preserving valid business records.

Cleaning Procedure

- Load the dataset with Pandas and inspect the first records using `head()`.
- Review data types, null counts, and memory structure using `info()`.
- Use `describe()` to inspect distributions, minimums, maximums, and unusual values.
- Check dataset dimensions using `shape()`.
- Remove exact duplicate records and investigate repeated order identifiers.
- Convert Order Date and Ship Date to datetime format.
- Remove or investigate records with missing shipment dates.
- Remove invalid records where Ship Date occurs before Order Date.
- Standardize State, Region, and Product Name text values.
- Verify that Sales, Cost, Units, and Gross Profit contain valid numeric values.
- Confirm that Gross Profit is logically consistent with Sales minus Cost where applicable.

Why Validation Matters

A cleaned dataset improves trust in every later result. If states are written in multiple formats, the same location may be counted as separate states. If dates are stored as text, monthly trends may be sorted alphabetically rather than chronologically. If duplicate rows remain, route volume and revenue may be overstated. These problems are not merely technical; they can lead directly to poor operational decisions.

Quality Output

The expected output is a clean, consistent dataset exported as `cleaned_shipping_data.csv`. This file becomes the single source used by the visualization layer and Streamlit application, reducing the risk that different dashboards use different versions of the data.

4. Feature Engineering

Feature engineering means creating new variables that express the business questions more clearly than the original fields. The raw dataset contains Order Date and Ship Date, but the most useful operational measure is the number of days between them. The raw product name is also less useful for logistics decisions until it is connected to a factory and customer destination.

Shipping Lead Time

Shipping Lead Time = Ship Date – Order Date. This value measures the elapsed number of days between order placement and shipment. It is the primary time-performance metric used throughout the project. Average lead time summarizes typical performance, while the distribution and standard deviation reveal inconsistency.

Route Construction

A route is defined at two levels: Factory → Customer Region and Factory → Customer State. The region-level route is useful for strategic planning, while the state-level route supports detailed diagnosis. For example, a factory may perform well across an entire region but struggle with one particular state because of distance, congestion, or local handling conditions.

Additional Features

- Order Month and Order Year for trend analysis.
- Shipping Days as a standardized duration field.
- Factory Name mapped from Product Name.
- Factory-to-State route label.
- Factory-to-Region route label.
- Delay flag based on a selected lead-time threshold.
- Optional efficiency score for ranking comparable routes.

Factory Coordinates

The project provides coordinates for four factories: Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory, with the listed latitude and longitude values. These coordinates can support geographic mapping and distance-oriented interpretation in the dashboard.

5. Exploratory Data Analysis

Exploratory Data Analysis, or EDA, is used to understand the shape of the data before formal conclusions are made. It combines summary statistics with visualizations so that analysts can identify patterns, outliers, unusual groups, and relationships between variables.

Analysis Areas

- Shipping lead-time distribution: identifies typical duration, spread, and extreme delays.
- Sales and gross profit: shows the commercial importance of products and routes.
- Units sold: reveals demand volume and possible fulfillment pressure.
- Ship mode distribution: shows how frequently each shipping option is used.
- Region-wise and state-wise orders: identifies geographic demand concentration.
- Factory-wise shipments: compares production sources and workload.
- Product division analysis: explains how chocolate, sugar, and other products contribute.
- Monthly order trend: detects seasonality and changes in demand.
- Correlation analysis: explores relationships among sales, units, cost, profit, and lead time.
- Outlier detection: flags unusually long or short shipments for investigation.

Recommended Visualizations

Histograms are appropriate for lead-time distributions. Boxplots compare lead-time spread across factories, regions, and ship modes. Bar charts rank routes and states. Line charts show monthly trends. Heatmaps can display factory-by-region performance, and geographic maps can show state-level bottlenecks. Matplotlib supports the static analytical visuals, while Streamlit makes the same insights interactive.

Interpreting EDA Carefully

A high average lead time does not always mean a route is operationally poor; it may reflect a small number of remote destinations or low shipment volume. Similarly, a high-volume route with a moderate average delay may deserve more attention than a very slow route with only a few orders. For this reason, volume and time performance should always be examined together.

6. Route Efficiency Analysis

Route efficiency analysis converts shipment records into route-level performance summaries. Each route is grouped by factory and customer geography, and the resulting groups are ranked according to speed, volume, consistency, and delay behavior.

Core Calculations

- Route volume = count of orders or shipment records assigned to the route.
- Average lead time = mean Shipping Lead Time for the route.
- Lead-time variability = standard deviation or another spread measure.
- Delay frequency = delayed shipments divided by total shipments, multiplied by 100.
- Fastest routes = routes with the lowest reliable average lead time.
- Slowest routes = routes with the highest average lead time or delay frequency.
- Route efficiency score = normalized score that supports comparison across routes.

Benchmarking Rules

The project requirements call for identifying the top 10 most efficient and bottom 10 least efficient routes. Rankings should include a minimum-volume rule where possible, because a route with one shipment cannot be compared fairly with a route containing hundreds of shipments. A practical leaderboard may show route name, factory, destination, order count, average lead time, variability, delay rate, and rank.

Efficiency Score Concept

A normalized efficiency score can be designed so that shorter lead times receive higher scores. One simple approach is to scale route performance between the best and worst observed values. The exact formula should be documented in the dashboard, and the score should be treated as a comparison aid rather than a replacement for the underlying lead-time metrics.

Bottleneck Identification

The most important bottlenecks are often high-volume routes with poor performance. These routes affect more customers and more orders than low-volume problem routes. A combined view of route volume, average lead time, and delay frequency allows management to prioritize improvement work according to business impact.

7. Geographic and Ship Mode Analysis

Geographic analysis explains where shipping performance differs across the network. Aggregating by region and state can highlight areas with longer lead times, greater variability, or higher delay frequency. A map makes these patterns easier to communicate because decision-makers can see the spatial distribution of performance.

Geographic Bottlenecks

A region with high average lead time may require transportation or warehouse review. A state with high order volume and poor performance may indicate capacity pressure, inefficient routing, or a mismatch between the fulfillment factory and customer location. Geographic results should be interpreted alongside factory workload and route volume rather than in isolation.

Ship Mode Comparison

Ship mode analysis compares the lead-time behavior of standard and expedited shipping, or any other modes present in the dataset. The dashboard can show average lead time, shipment count, delay frequency, and variability for each mode. The goal is descriptive: faster service may improve delivery time, but the business must also consider cost and availability.

Cost-Time Tradeoff

The project does not require a causal cost model, so the findings should not claim that one ship mode is universally better. Instead, the analysis should identify where faster modes appear to reduce delays and where their use may be justified, especially for delay-prone routes or high-priority customers.

Interactive User Controls

- Date range filter for analyzing selected periods.
- Region and state selectors for geographic drill-down.
- Ship mode filter for focused comparisons.
- Lead-time threshold slider for defining delays.
- Route and order-level tables for detailed investigation.

8. Tableau Dashboard

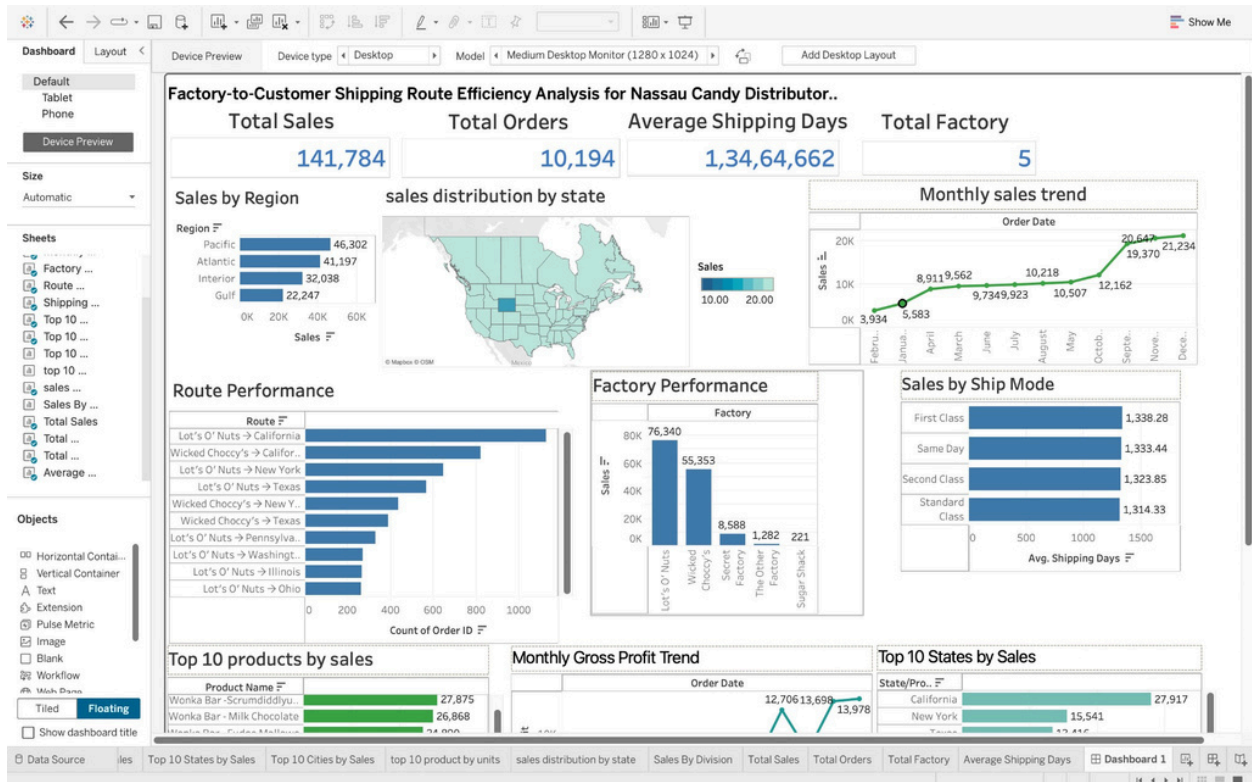


Figure 1. Supplied Tableau dashboard overview.

KPI cards

The Tableau view contains Total Sales, Total Orders, Average Shipping Days and Total Factory. The screenshot shows 141,784 sales, 10,194 orders, 5 factories and an extremely large displayed shipping-day number that should be validated.

Sales by Region

Pacific is the largest visible region, followed by Atlantic, Interior and Gulf. Use this as a commercial concentration view and cross-check it with logistics performance.

Sales Distribution by State

The map provides geographic context. Use it to discover concentration, then use a ranked table for exact comparisons.

Monthly Sales Trend

The line chart shows time movement and supports capacity planning.

Route Performance

Factory-to-customer routes are ranked for commercial importance. Route volume and shipping time should be considered together.

Factory Performance

The visible factory values are approximately 76k, 55k, 8.6k, 1.3k and 220. This reveals strong workload concentration.

Sales by Ship Mode

The screenshot compares First Class, Same Day, Second Class and Standard Class using displayed shipping-day values. Metric definitions should be documented before operational use.

Top Products

Wonka Bar variants are prominent in the visible product ranking.

Monthly Gross Profit

The line chart shows monthly profit movement with peaks and dips.

Top States

California is the leading visible state, followed by New York and Texas.

9.Streamlit Dashboard

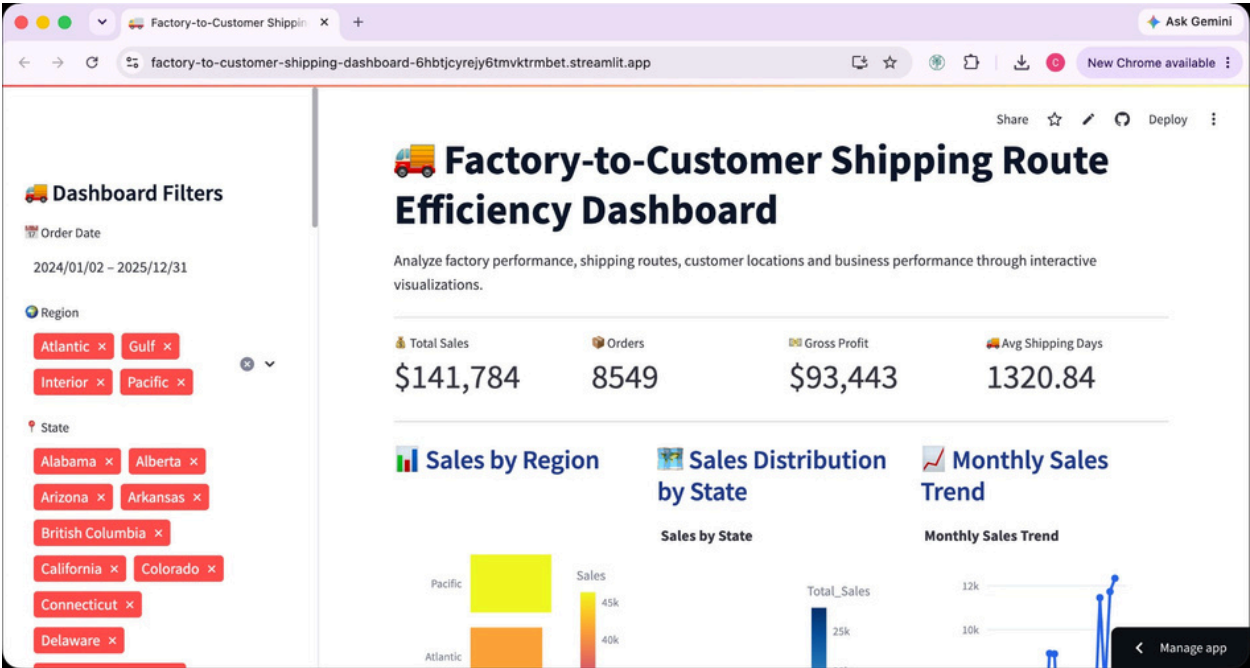


Figure 2. Streamlit opening view with filters and KPI cards.

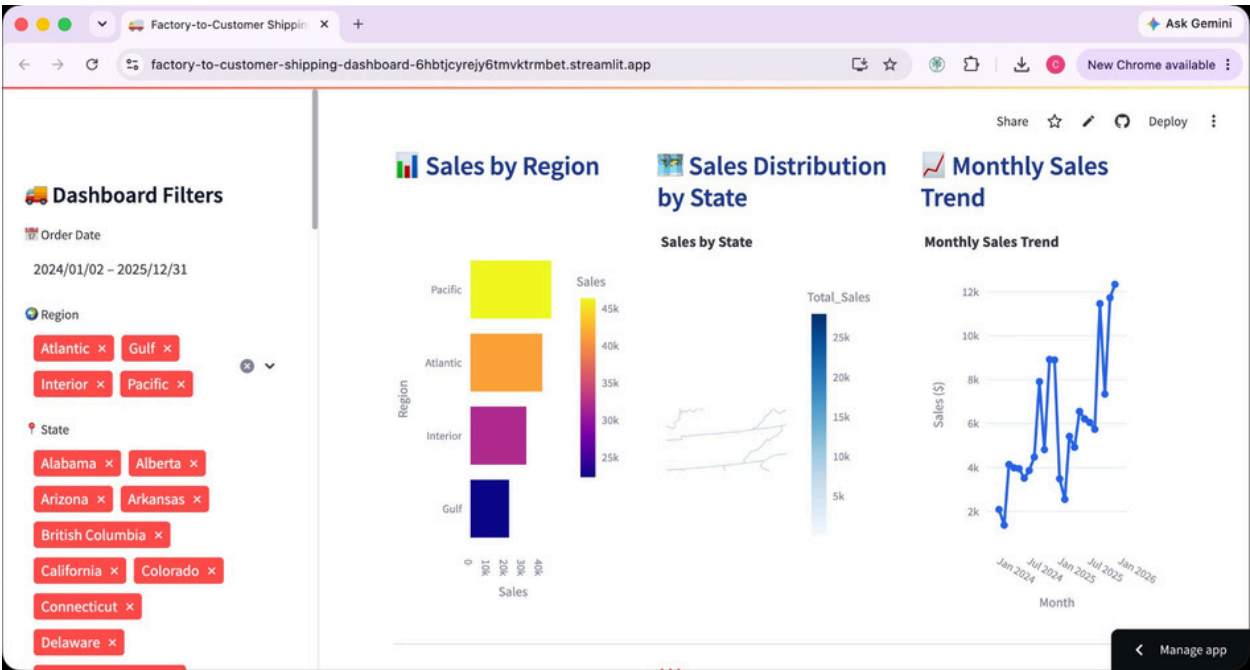


Figure 3. Streamlit regional, state and monthly overview.

The dashboard highlights strong Pacific-region sales, lower Gulf performance, varied state-level sales, and fluctuating monthly trends, with sales strengthening toward the end, supporting better regional planning and resource allocation.

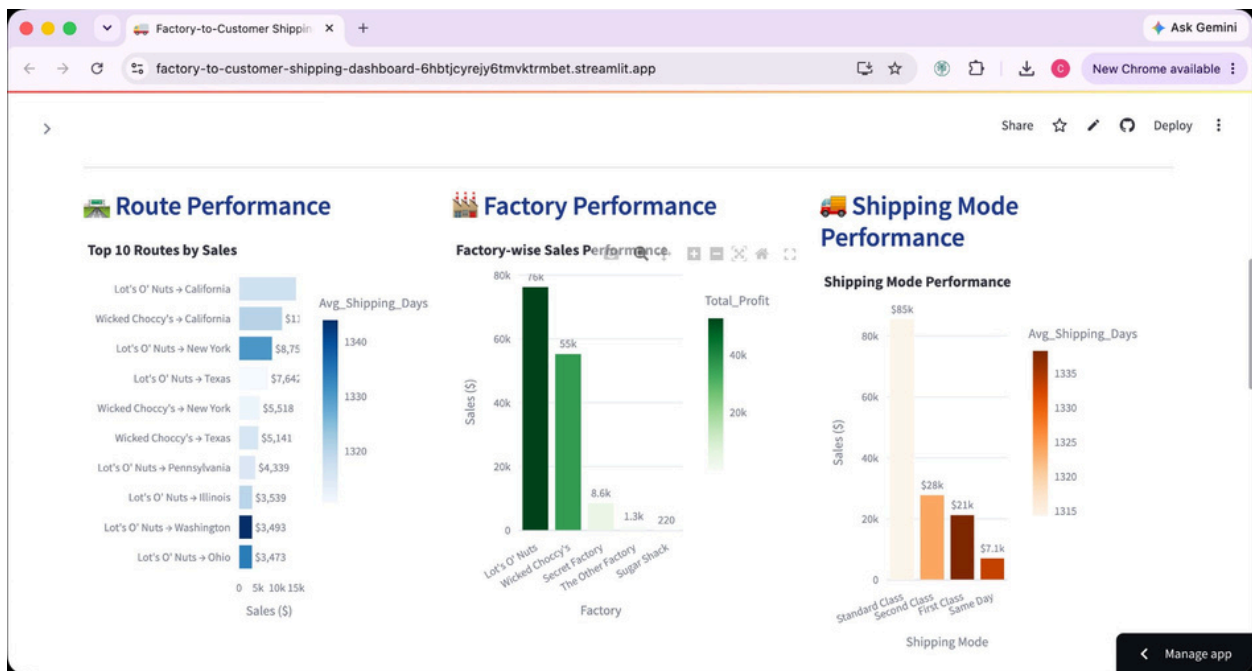


Figure 4. Streamlit route, factory and shipping-mode performance.

The dashboard highlights route, factory, and shipping performance, with Lot's O' Nuts leading factory sales, California routes performing strongly, and Standard Class generating the highest sales among shipping modes.

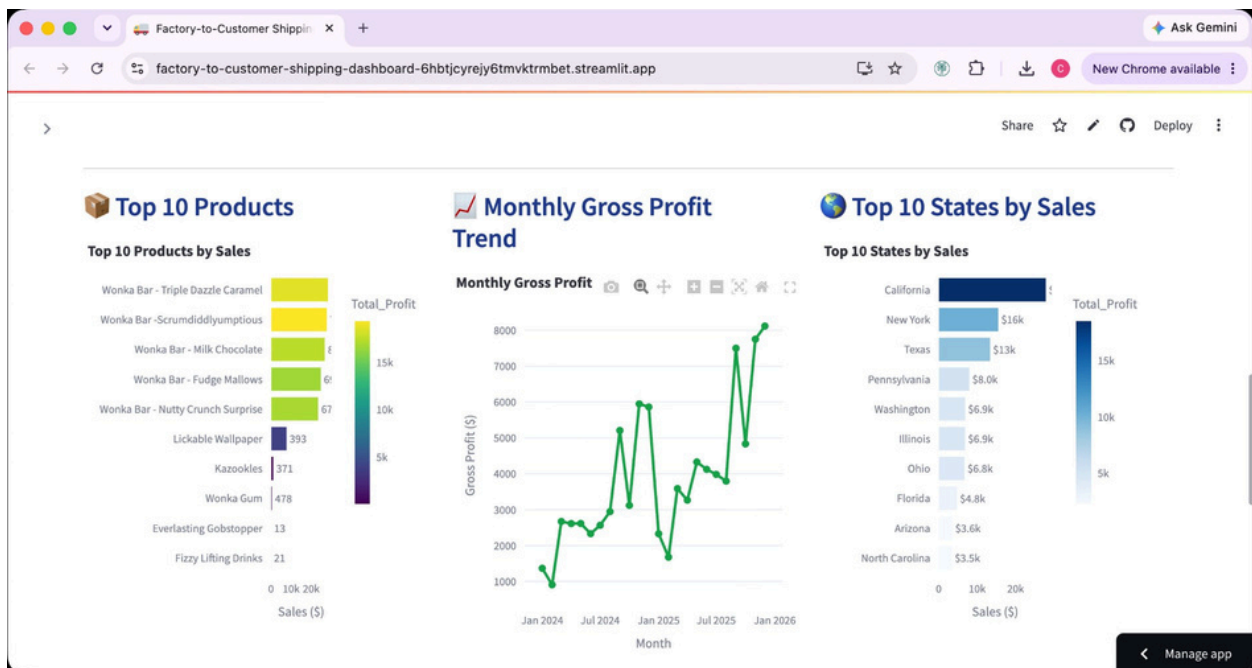


Figure 5. Streamlit products, gross profit and top states.

The dashboard highlights top-performing products, states, and profit trends, with Wonka Bar products leading sales, California recording the highest sales, and gross profit showing stronger performance toward the period's end.

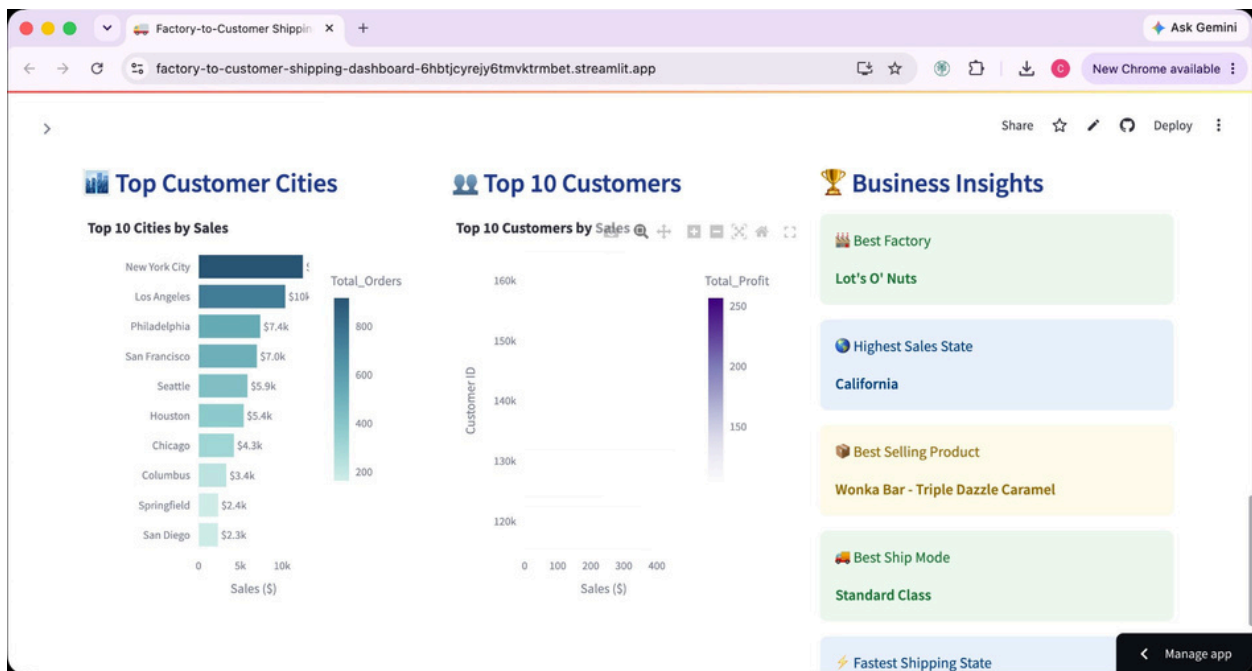


Figure 6. Streamlit customer cities, customers and business insights.

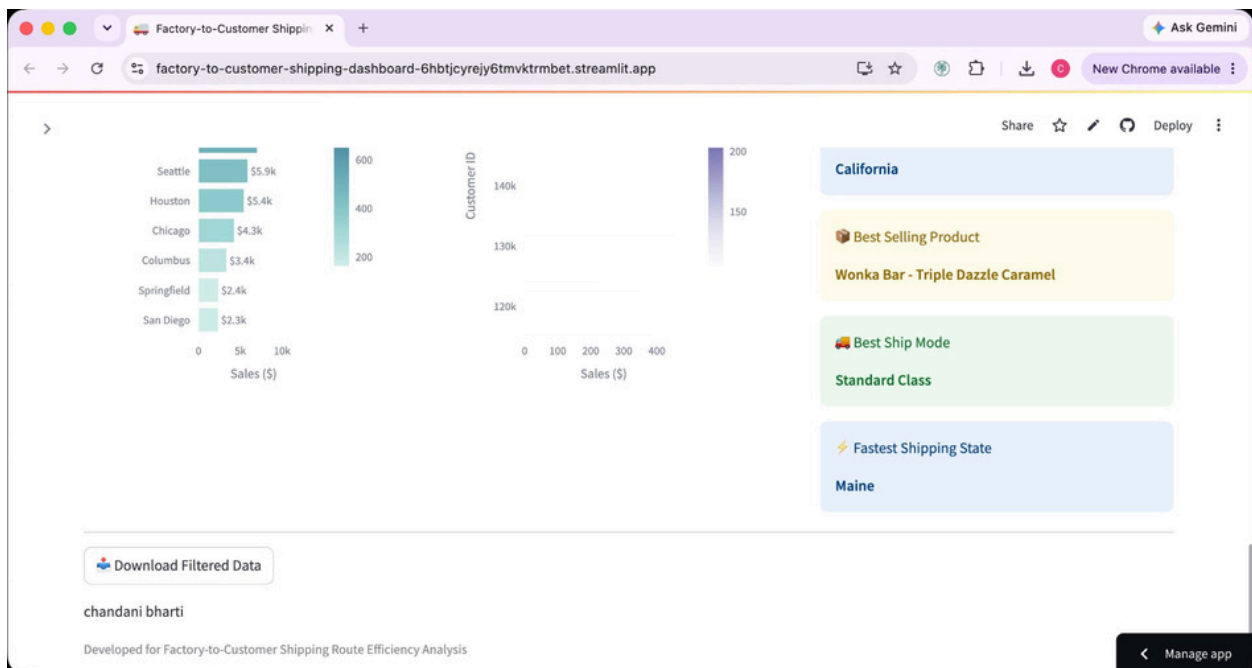


Figure 7. Streamlit business insights and footer.

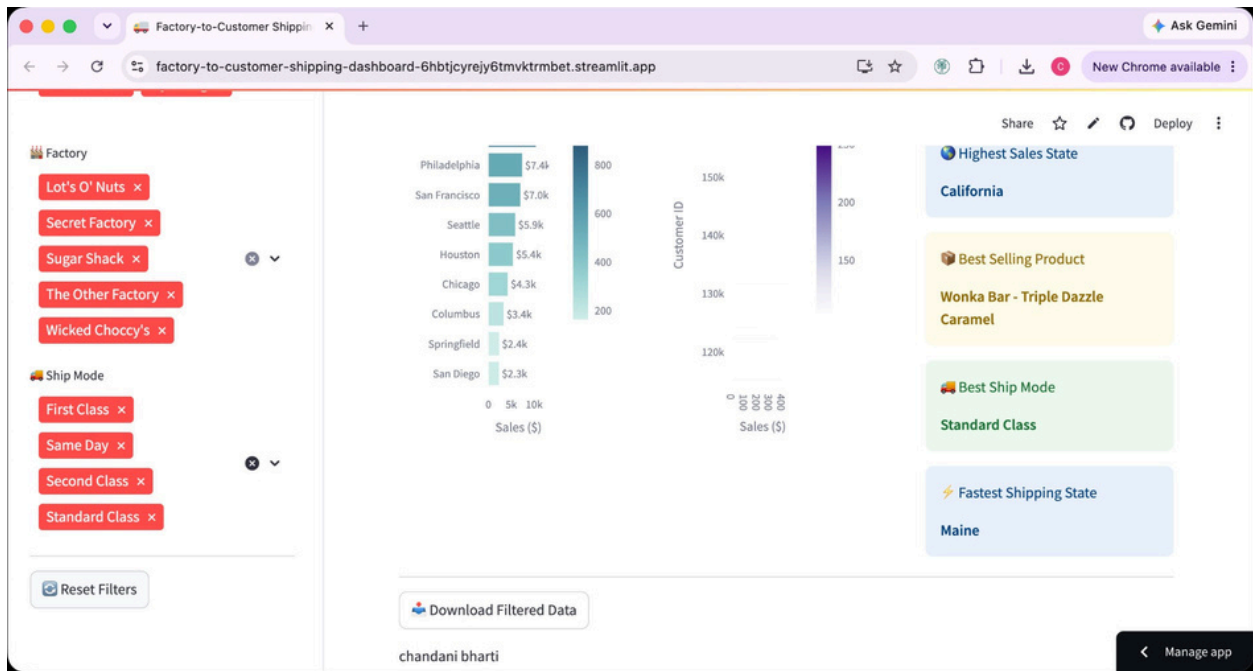


Figure 8. Streamlit factory/ship-mode filters and filtered-data download.

The dashboard provides a comprehensive view of factory-to-customer shipping and business performance. The analysis indicates that the Pacific region generates the highest sales, while Gulf has comparatively lower sales. Sales fluctuate significantly over time, with stronger performance toward the later period. New York City and Los Angeles represent the most valuable customer cities, while California is the highest-sales state. Lot's O' Nuts is identified as the best-performing factory, Wonka Bar – Triple Dazzle Caramel as the best-selling product, Standard Class as the best-performing shipping mode, and Maine as the fastest-shipping state. Overall, these insights can help management optimize regional operations, prioritize high-value markets, improve shipping decisions, and identify opportunities for growth

10. Business Insights and Recommendations

The project produces actionable insights rather than only descriptive charts. The most useful output is a prioritized list of routes and locations where operational changes could improve delivery reliability.

Expected Business Insights

- Fastest factory-to-customer routes can serve as internal benchmarks.
- Slow routes and high delay-frequency routes should receive diagnostic attention.
- High-volume states with poor performance represent the largest immediate improvement opportunity.
- Factory-level comparisons can reveal workload imbalance or geographic mismatch.
- Monthly trends can support capacity planning during high-demand periods.
- Ship mode comparisons can guide targeted use of expedited services.
- Product-to-factory mapping can show whether certain divisions are associated with specific geographic patterns.

Recommendations

- Review consistently slow routes for distance, carrier, handling, and scheduling causes.
- Use expedited shipping selectively on routes with repeated delays or high customer impact.
- Evaluate warehouse or factory allocation for regions with high volume and weak performance.
- Set a documented lead-time threshold and monitor delay frequency continuously.
- Create a monthly logistics review using the Streamlit KPI dashboard.
- Investigate outliers individually rather than automatically removing them.
- Recalculate route rankings after operational changes to measure improvement.

Implementation Priorities

The first priority should be high-volume bottlenecks because improvement there can affect the greatest number of shipments. The second priority should be routes with high variability, since inconsistency makes planning difficult. The third priority should be low-volume extreme delays, which may indicate special cases, data quality issues, or remote delivery constraints.

11. Limitations, Conclusion, and Deliverables

Limitations

The provided project summary documents the workflow and intended analyses but does not provide the final row count, exact calculated averages, threshold value, route leaderboard, or dashboard screenshots. Therefore, this paper explains the analytical design and defensible interpretation without fabricating numerical findings. Final numeric claims should be inserted after running the completed notebook or Streamlit application on the source dataset.

Route efficiency is also influenced by factors that may not exist in the dataset, such as carrier identity, transportation distance, weather, warehouse processing time, holidays, and customer priority. The analysis should therefore be interpreted as operational evidence from shipment records, not as proof of a single causal factor.

Conclusion

This project establishes a complete data-driven framework for understanding factory-to-customer shipping efficiency at Nassau Candy Distributor. Through cleaning, feature engineering, EDA, route aggregation, geographic analysis, and dashboard development, raw data becomes a practical decision-support resource. The combined use of Python, Pandas, NumPy, Matplotlib, Excel, VS Code, and Streamlit supports both technical reproducibility and business usability.

Deliverables

- Research paper covering EDA, route analysis, insights, and recommendations.
- Cleaned dataset exported as `cleaned_shipping_data.csv`.
- Streamlit web application with filters, KPIs, maps, charts, and drill-down views.
- Executive summary suitable for management or government stakeholders.
- Supporting code and documentation developed in VS Code.

Source Basis

This expanded paper is based on the user-provided Nassau Candy project requirements and the uploaded executive summary, including the documented workflow, dataset fields, factory coordinates, product-factory mapping, KPIs, dashboard modules, and recommendations [file:1].